

Modeling of multi-photon events of a superconducting single-photon nanostripe detector for photon number resolution experiments

Introduction

Superconducting nanostripe single photon detectors (SNSPDs) [1] exhibit near-unit quantum efficiency [2,3], gigahertz counting rates [4], picosecond jitter [5], sub-hertz dark counts [6], and a relatively high operating temperature (2-4 K). Their operation [1] is schematically shown in FIG. 1.

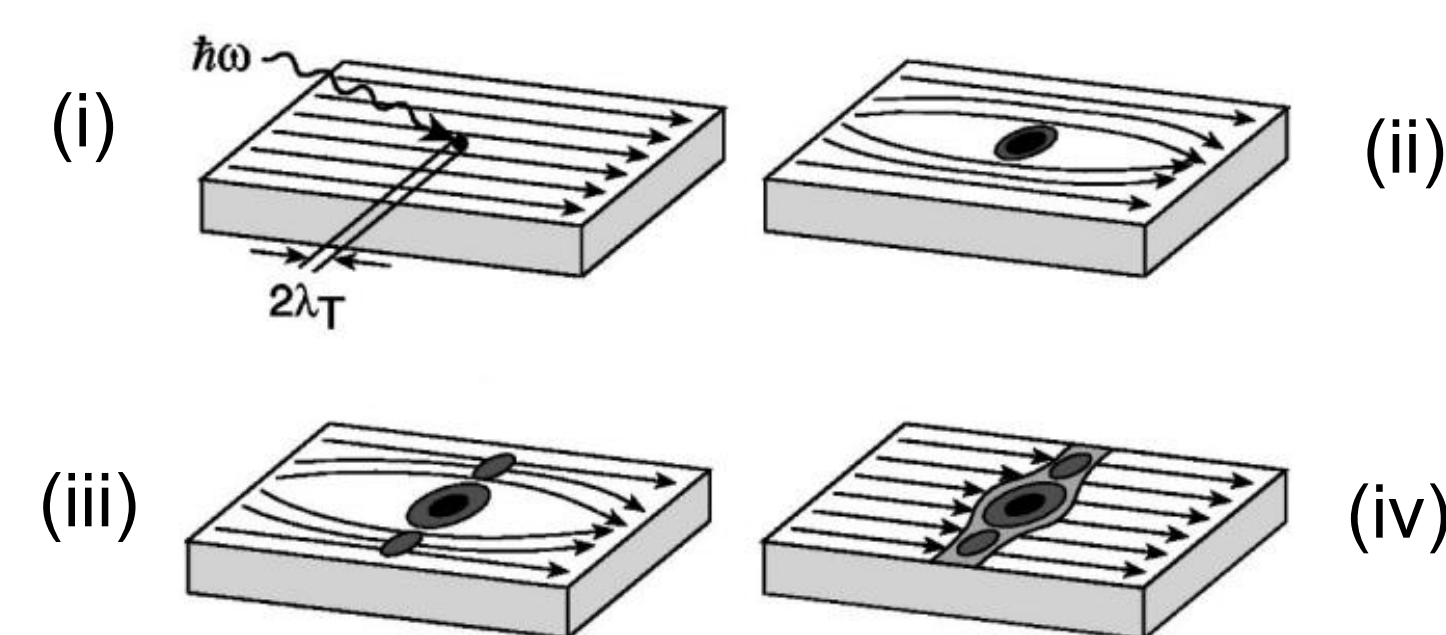


FIG. 1: Schematics of SNSPD photon detection.

(i) The nanostripe is maintained at a temperature far below its critical temperature T_c and biased with a dc current I_b just below the critical current I_c . (ii) After a photon of energy is absorbed, a hotspot is created. (iii) The hotspot expands, forcing I_b to flow along its periphery. (iv) As the hotspot continues to grow, a resistive barrier is formed across the width of the nanostripe.

Despite the above listed advantages, the SNSPD operates primarily in threshold mode, and photon number resolution (PNR) is challenging. We worked on using the single-pixel or intrinsic approach to solve the SNSPD readout concept, where PNR is achieved by either discrimination of SNSPD waveform amplitude or its rise time. Tracing the rising-edge-time is the most common photonic number resolving technique for the intrinsic approach.

Schematic Diagram of the Dynamic Model of SNSPD

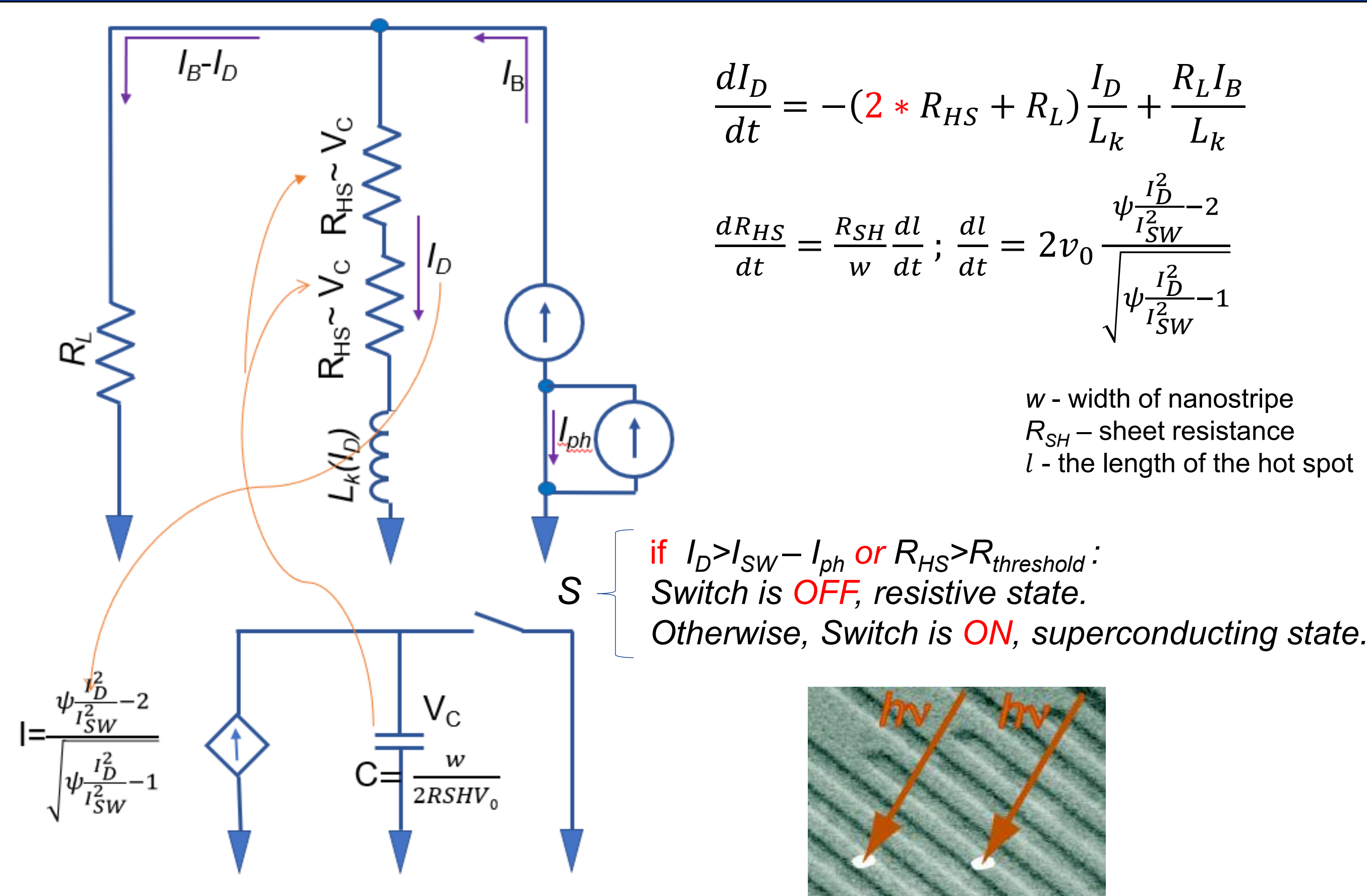


FIG. 2: Schematic diagram of the Dynamic Model of SNSPD for the case of a two-photon absorption event.

We use a modified SPICE dynamic model [7] for the photoresponse simulation of up to five photon absorption events based on equivalent number hot spot resistances connected in series.

SPICE Simulation Results

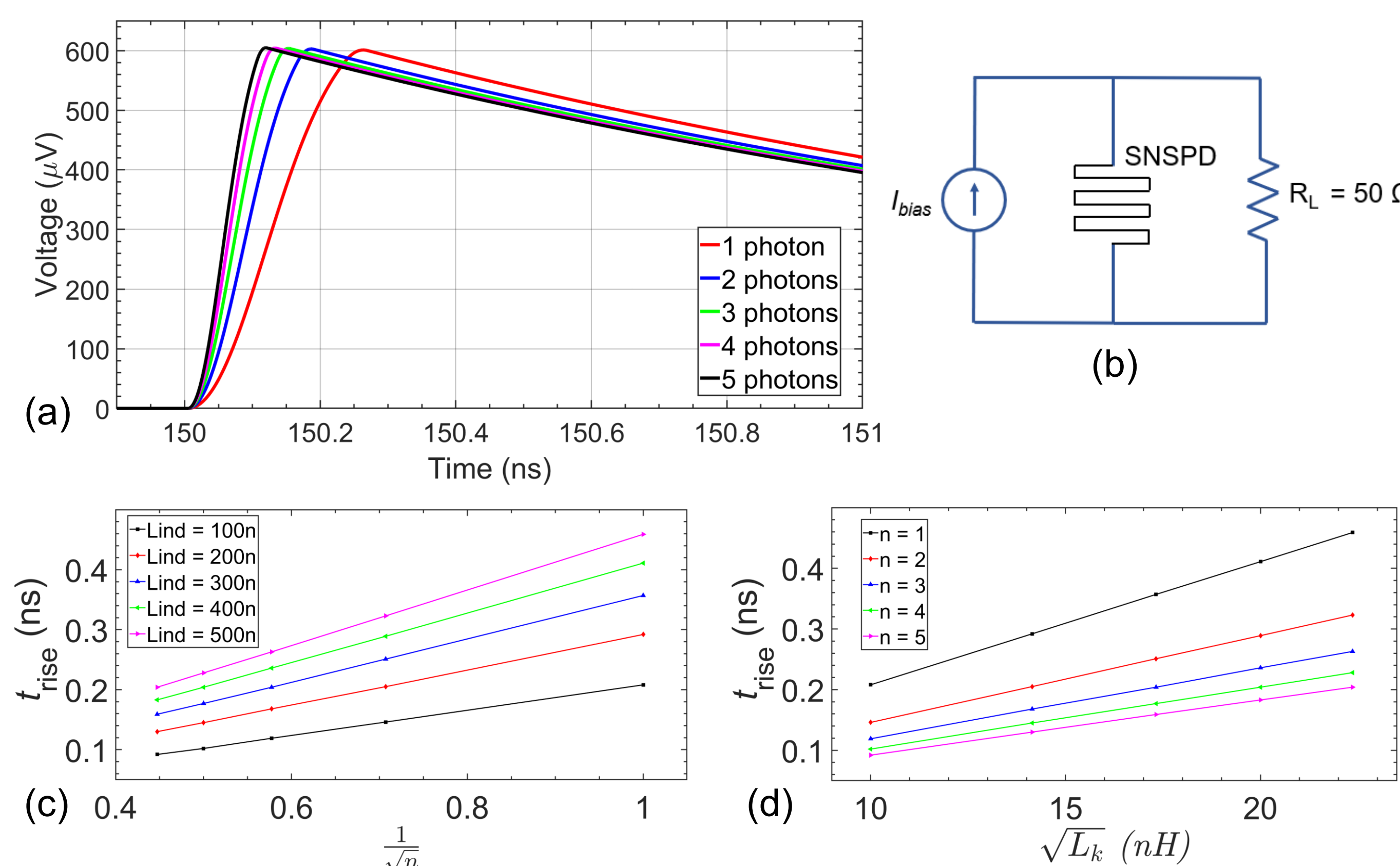


FIG. 3: (a) Plot of SNSPD pulses for 1-5 photon detection events, from (b) voltages across R_L in the simplify photoresponse electrical model of the SNSPD. Plot of rise time in relation with the number of photons (c), with the length or inductance of the nanostripe (d).

The total resistance of the nanostripe due to the presence of n hot spots is given by nR_{HS} . Under the electrothermal model of SNSPDs, the output pulses of the model shows the rise time of the pulse encodes the photon number as well as the length of the nanostripe with scaling $t_{rise} \propto \sqrt{\frac{L}{n}}$.

Digitization and Fitting of Smirnov Data

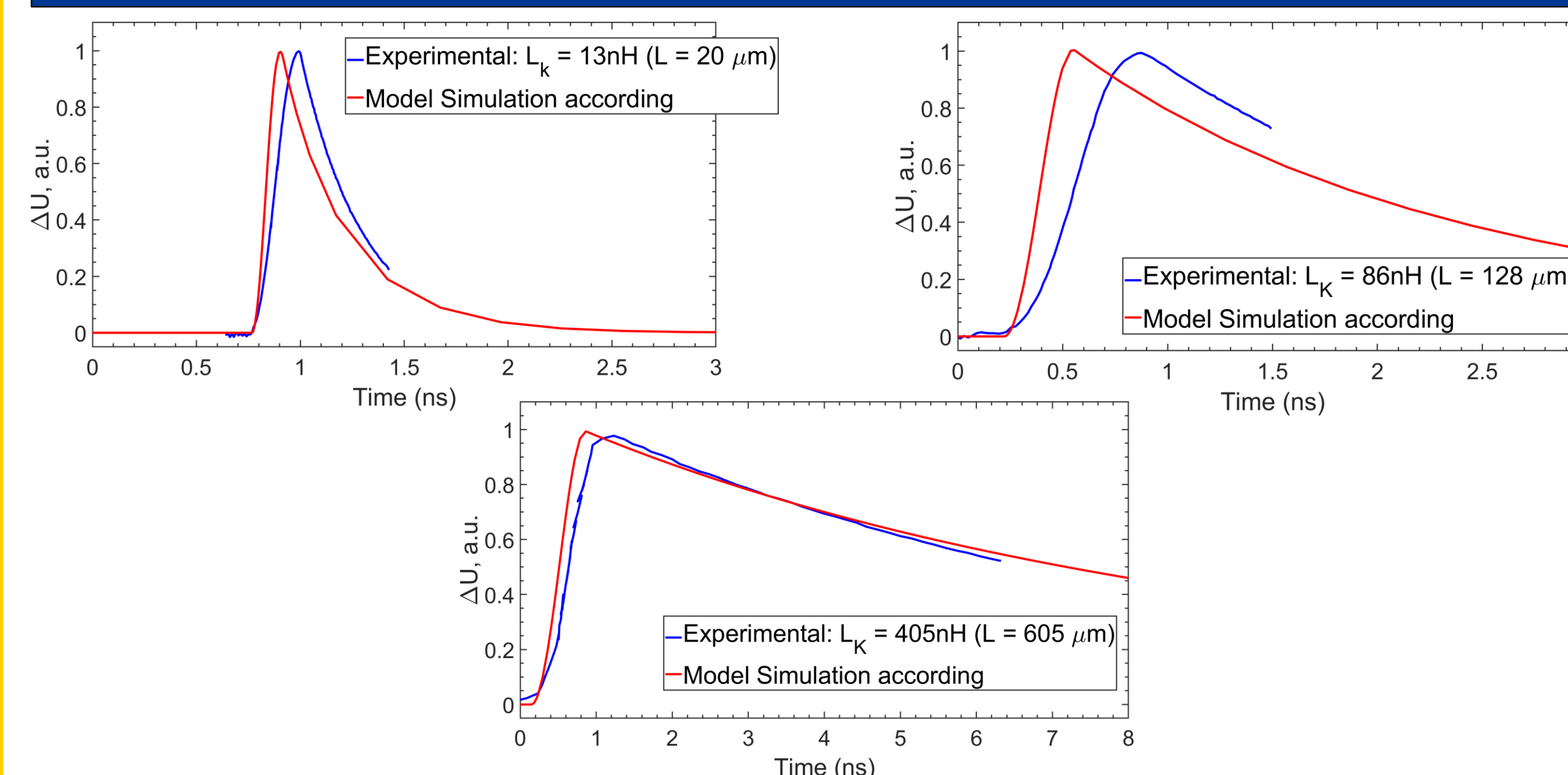


FIG. 4: Plots of model voltage pulses (red line) in comparison to experimentally measured pulses (blue line), for varied kinetic inductances.

The current model exhibits some discrepancies (faster t_{rise}) with single-photon experimental results, as it does not include jitter effects, microscopic hot-spot creation processes, or post-detection signal delays [7]. In addition, further refinement is also needed to investigate bias-current-dependent rise time events.

Photon number resolution using SFQ Readout Circuit

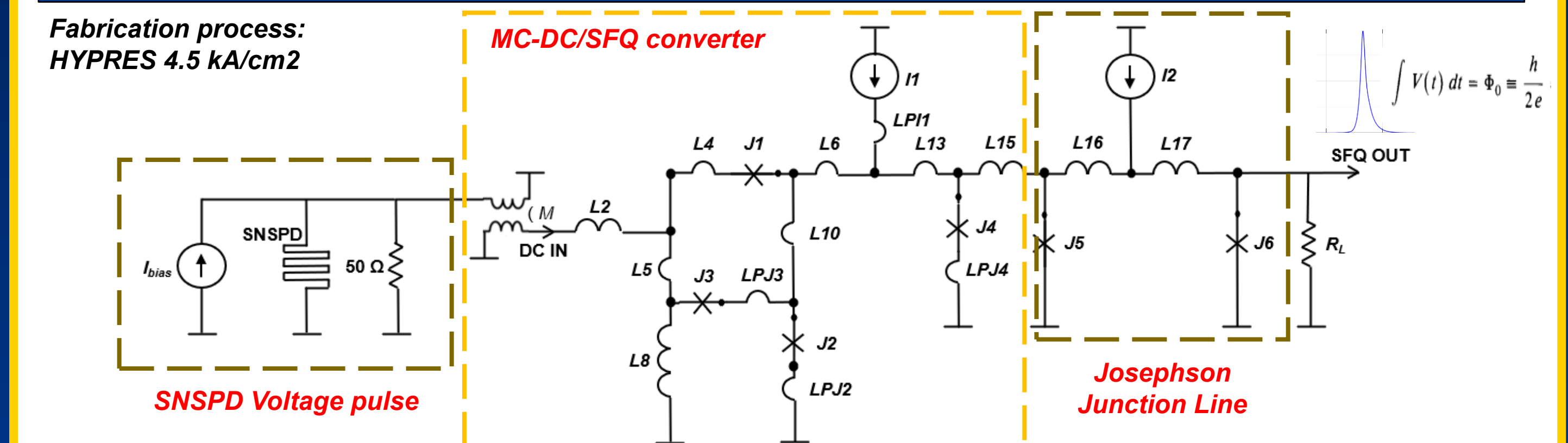


FIG. 5: Schematics for Single Flux Quantum (SFQ) readout circuit.

Future work will integrate this SPICE model with SFQ circuits for PNR using the PSCAN2 software (Superconductor Circuit Simulator).

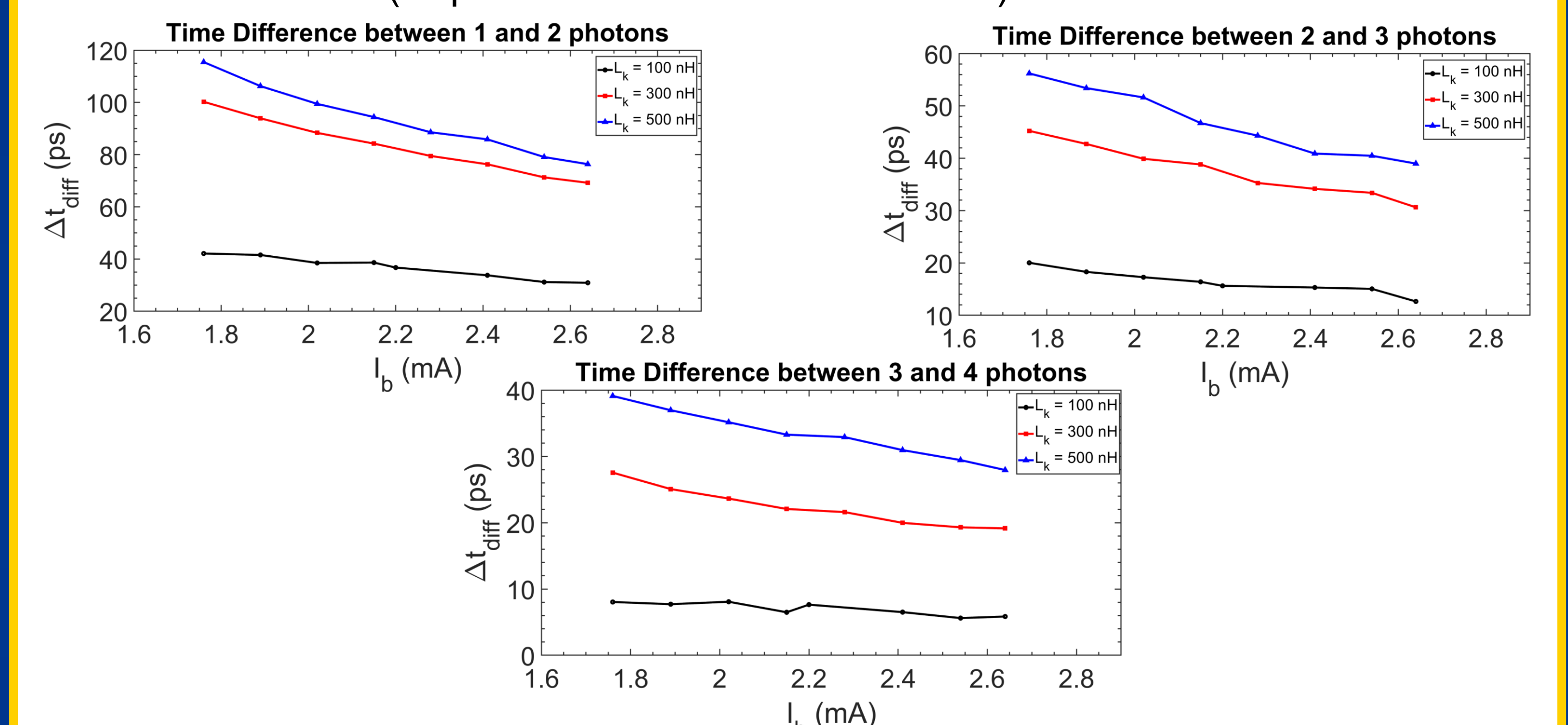


FIG. 6: Plots of time differences between SFQ rising edge times for n and $n + 1$ photons in relation with bias currents, for varied kinetic inductances.

Adjusting the kinetic inductance of SNSPD makes it possible to tolerate the time delay between SFQ pulses corresponding to different photon number events, which opens the possibility to design a SFQ readout system for PNR application.

Conclusions

Full dynamic model for a SNSPD that can describe dynamics of multiple optical-photon detection events has been implemented using the SPICE platform. This model can be further integrated as a device symbol into a complex circuit such as a readout circuit for verification and optimization.

References

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